



MinersBuddy

DGMS EXAM PREPARATION

COMPLETE EXAM SYLLABUS

BLASTER

UNRESTRICTED

Metalliferous Mines | Opencast and underground workings

Common syllabus: Mining Mate (Unrestricted)

13 + 8

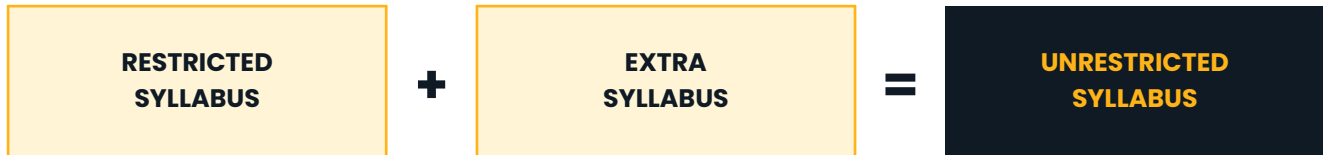
Core + Additional modules

198

Topics

Syllabus at a Glance

The Unrestricted syllabus covers opencast and underground workings. It contains the complete Restricted syllabus (Part A) plus the additional underground topics (Part B). The same syllabus applies to both Mining Mate and Blaster.



NO.	MODULE	CORE	ADDITIONAL
1	Mine Legislation	16	+ 5 (B1)
2	Duties, Reports & Examinations of Mate and Blaster	9	-
3	Opencast Mining Practice & Safety	23	-
4	Explosives, Blasting & Shotfiring (Opencast)	55	+ 18 (B2)
5	First Aid & Emergency Management	15	-
6	Miner's Diseases & Dust Hazards	7	+ 2 (B7)
7	Mine Plans & Sections	9	+ 11 (B6)
8	Mine Machinery (Surface and Opencast Basics)	4	+ 12 (B5)
9	Traffic Rules in Mines	1	-
10	Mine Working & General Safety	3	-
11	Methods of Working: Opencast	1	+ 1 (B4)
12	Mine Lighting	1	+ 1 (B8)
13	Abbreviations Used in Mining	1	-
B3	Precautions against Fire, Explosion & Inundation	-	+ 3 (B3)
Total		145	+ 53

How to use: Part A is the base you must master first. Part B adds underground topics that are examined only in the Unrestricted paper.

PART A

Restricted Syllabus (Core)

Opencast workings. This part is identical to the Restricted syllabus.

1 Mine Legislation

Acts, Rules & Circulars

- The Mines Act, 1952
- The Mines Rules, 1955 (health, sanitation, leave)
- Mines Vocational Training Rules, 1966
- DGMS Circulars (technical and general), e.g. Tech. Circular 01/2024 on electronic detonators in opencast mines

Metalliferous Mines Regulations, 1961 (core chapters)

- | | |
|---|---|
| ■ Preliminary | ■ Means of access and egress |
| ■ Returns, notices and records | ■ Transport of men and materials: haulage |
| ■ Examinations and certificates of competency and fitness | ■ Mine workings |
| ■ Inspectors and mine officials | ■ Explosives and shotfiring |
| ■ Duties and responsibilities of workmen, competent persons and officials | ■ Machinery and plant |
| ■ Plans and sections | ■ Miscellaneous and Schedules |

2 Duties, Reports & Examinations of Mate and Blaster

- | | |
|---|---|
| ■ Application form for examination | ■ Examinations / inspections by Mining Mate |
| ■ Details of required experience | ■ Precautions for flammable and harmful gases |
| ■ Subjects for Mate and Blaster examination | ■ Report of Mate |
| ■ Duties and responsibilities of a Blaster | ■ Report of Blaster |
| ■ Duties and responsibilities of a Mate | |

3 Opencast Mining Practice & Safety

Opencast fundamentals

- | | |
|--|-------------------------------------|
| ■ Opencast (open) mines | ■ Arrangement of footpaths in mines |
| ■ Advantages and disadvantages of opencast mines | ■ Height and width of benches |
| ■ Mine excavation / digging | ■ Spoil / debris dumps |

Statutory safety provisions

- | | |
|---|--|
| ■ Laws for the safety of workers in mines | ■ Rules for movement of tubs / mine cars |
| ■ Laws for fencing and doors / gates | ■ Dangers of surface railway wagons |
| ■ Rules for haulage roadways | ■ Fencing around opencast mines |
| ■ Rules regarding travelling roadways | ■ Roads for trucks and dumpers |

Operations and machines

- Supervision of work and maintenance of machines
- Precautions in drilling
- Precautions during blasting
- Precautions on opencast machines
- Safety functions of truck / dumper operators

Types of fencing

- S-type fencing: S1, S2, S3
- G-type fencing: G1, G2, G3, G4, G5
- U-type fencing: U1, U2
- Types of opencast fencing and their use

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Explosives, Blasting & Shotfiring (Opencast)

Explosives and detonators

- Explosives and their types
- Types of permitted explosives
- Explosives for opencast mines
- Liquid explosives (slurry / emulsion)
- Detonator (cap); ordinary detonator: advantages and disadvantages
- Electric detonator and delay detonator
- Detonating relay
- Detonating fuse (Cordtex)
- Types of fuse joints

Statutory rules, storage and handling

- Appointment of Shotfirer and Blaster
- Conditions and precautions in using permitted explosives
- Blasting materials / equipment
- Magazine and precautions in magazine
- Rules regarding explosives in mines
- Safety instructions for shotfirer / blaster helpers
- Place of shelter
- Safety instructions for explosive carriers
- Transit slip

Charging and firing

- General rules for drilling, charging, stemming and firing
- Exploder / blasting machine
- Stemming clay / material
- Methods of firing; firing with electric detonator
- Types of firing circuits and testing of circuit
- Firing of hole
- Zapro ohmmeter circuit tester

Misfires

- Causes of and precautions against misfire
- Misfire in electric detonator
- Rules in the regulations for removing or extracting misfires
- Misfires in large diameter blasting
- General precautions

Opencast blasting practice

- Blasting in opencast mines
- Quantity of explosives
- Methods of blasting; method of blasting in opencast
- Deck charging (charging at different heights)
- Protection from flying rocks during blasting
- Work and precautions in opencast over underground mine
- Secondary blasting
- Dangers and causes of explosive accidents in mines
- Appendices 1 to 6

Advanced opencast blasting (reference: L. C. Kaku, Part Two)

- Principles, requirements and importance of blasting in opencast mines
- Practical aspects of explosive quality
- Methods of initiating explosives; sequential blasting machine
- Bulk transport of explosives
- Deep hole blasting and bulk mix explosive blasting
- Blast design: drilling pattern, firing pattern, delay interval, cord relay
- Special methods: sleeping hole, blasting in fire area, air deck, ANFO, cast blasting, pilot hole
- Controlled blasting: pre-splitting, smooth wall blasting, line drilling
- Control of fly rock, ground vibration and air blast; muffle blasting
- Nonel / shock tube system and electronic detonator
- Safety fuse blasting
- Blasting record and daily report of blasting overman
- Main causes of accidents from blasting in opencast mines
- Explosives Rules, 2008: an introduction
- Purchase of explosives with powder factor guarantee
- Duties of Magazine In-charge

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First Aid & Emergency Management

- Section A: Basic first aid techniques (CPR, recovery position, assessment)
- Section B: Respiratory system and respiration (choking, asthma, suffocation)
- Section C: Heart, blood circulation, shock (bleeding, wounds, amputation)
- Section D: Bones, joints and muscles (fractures, dislocations, sprains)
- Section E: Nervous system and unconsciousness (head injury, stroke, seizures)
- Section F: Gastrointestinal tract, diarrhoea, food poisoning and diabetes
- Section G: Skin, burns, fever, heat exhaustion, heatstroke and hypothermia
- Section H: Poisoning (ingestion, inhalation, injection)
- Section I: Bites and stings (animal, snake, insect)
- Section J: Senses (eyes, ears, nose, skin) and swallowed foreign objects
- Section K: Urinary and reproductive systems; emergency childbirth
- Section L: Psychological first aid
- Section M: Specific emergency situations and disaster management (triage)
- Section N: First aid techniques (dressings, bandages, stretchers, transport)
- Appendices: first aid kits and references

6 Miner's Diseases & Dust Hazards

- Measurement of dustiness
- Dust control by the respiratory system
- Pneumoconiosis: classification of inorganic dusts, types of dust diseases
- Permissible dust concentrations
- Prevention and treatment of dust diseases
- Dust sampling methods and instruments
- Ailments of the ear (noise hazards)

7 Mine Plans & Sections

- Surface plan
- Geological plan
- Water danger plan
- Joint survey plan
- Electrical plan
- Combined / composite plan
- Manpower distribution plan
- Accident plan
- Conventions and symbols for preparing plans and sections

8 Mine Machinery (Surface and Opencast Basics)

- Wire ropes: composition, classification, flexibility and lay, constructions, selection and maintenance
- Belt conveyors
- Aerial ropeways: types, applicability, bi-cable ropeway components
- Mine pumps: centrifugal, turbine and other types

9 Traffic Rules in Mines

- Statutory traffic and transport guidelines in mines

10 Mine Working & General Safety

- Risk assessment
- Safety committees
- General mine safety

11 Methods of Working: Opencast

- Opencast extraction methods

12 Mine Lighting

- Mine illumination standards

13 Abbreviations Used in Mining

- Standard technical acronyms and definitions

PART B

Additional Syllabus (Unrestricted only)

Extra underground topics added on top of the Restricted syllabus.

B1 Mine Legislation: Additional Provisions

Metalliferous Mines Regulations, 1961 (additional chapters)

- Ladders and ladderways
- Transport of men and materials: winding in shafts
- Precautions against dangers from fire, dust, gas and water
- Ventilation
- Lighting and safety lamps

B2 Underground Explosives & Shotfiring

Reference: L. C. Kaku, Part One (Underground)

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|---|---|
| ■ Information about explosives | ■ Misfire, socket, blown through, blown out and cut-off |
| ■ Cap or detonator | ■ Some special points; card system |
| ■ Magazine | ■ Hole pattern |
| ■ Carrying and storing explosives in the mine | ■ Air deck solid blasting |
| ■ Shotfirer's equipment | ■ Ring-hole blasting in Blasting Gallery panel |
| ■ Making a priming cartridge | ■ Blasting in stone drift and sinking shaft |
| ■ Charging and stemming the hole | ■ Method of writing report |
| ■ Methods of firing | ■ Duties and responsibilities of Shotfirer |
| ■ Solid blasting | ■ Accidents from shotfiring in underground mines |

B3 Precautions against Fire, Explosion & Inundation

- Management of mine fires
- Firedamp and mine explosions
- Water danger and inundation

B4 Methods of Working: Underground

- Underground extraction methods

B5 Mine Machinery: Underground Haulage & Winding**Haulage**

- Rope haulage: direct rope, double drum balanced haulage and others
- Haulage safety devices
- Locomotive haulage
- Scraper chain conveyors and scraper haulage

Winding and shaft

- Winding systems: system comparisons
- Headgear and pulleys
- Cages vs. skips
- Shaft fittings and safety devices
- Pit-top and pit-bottom layouts
- Man-riding and material transport systems

Face machinery and handling

- Coal face machineries: LHD, shuttle car, gathering arm loader
- Coal handling plant (CHP)

B6 Mine Plans & Sections: Underground

- Underground plan
- Vertical mine sections
- Ventilation plan and section
- Combined section / sections
- Rescue plan / tracing
- Abandoned mine plan
- Stone dusting plan
- Sampling (zoning) plan
- Fire-fighting plan
- Subsidence plan and sections
- Sketch plan for systematic support rules

B7 Miner's Diseases: Additional

- Nystagmus
- Ankylostomiasis

B8 Safety Lamps & Underground Lighting

- Safety lamps

Reference Material

REFERENCE	SOURCE / AUTHOR
Mate and Blaster Goshti (Metal Mine Mining Examinations)	Arvind Kumar
Mine Explosives and Practical Blasting Techniques (Underground and Opencast)	L. C. Kaku
Mine Machinery: A Comprehensive Guide	Prof. Devidas S. Nimaje, NIT Rourkela
The Mines Act 1952, Mines Rules 1955, Metalliferous Mines Regulations 1961, Mines Vocational Training Rules 1966	Statutory texts
First Aid and Emergency Management manual; Miner's Diseases and Dust Hazards; Mine Plans and Sections	MinersBuddy content repository